

STEP 7

Streamlit user interface

A simple app for non-technical users, built on top of the API.

Tabs

Tab	What it shows
Predict	Form with date, business type, food category, quantity, demand, shelf life, temperature, weather and promotion → predicted level with colour, advice, probability chart, and 'Served by' + model version
Model	Best model, version, rows, quality gate, scores of all three candidates, confusion matrix
Monitoring	Run a drift check on a normal or heatwave batch and see which features drifted

How it talks to the model

The app sends the form to `API_URL/predict` (default `http://localhost:8000`, or `http://api:8000` inside docker compose). If the API cannot be reached, it falls back to loading the model file directly, and the caption changes to *'local model (API offline)'*, so the app still works for a demo.

Run it

```
streamlit run app/streamlit_app.py # http://localhost:8501
```

Advice shown for each level

Level	Advice
Low	Order looks right for expected demand.
Medium	Some waste likely - consider a smaller order or a promotion.
High	High waste risk - cut the order, run a promotion or plan a donation.

Watch out

Port 8501 can only be used by one app at a time. If an older Streamlit app (for example the IMDB project) is still running, stop it with Ctrl+C in its terminal, or start the other app with `--server.port 8502`.